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Demographic modeling to fine-tune conservation targets: importance of pre-adults for the decline of an endangered raptor

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Abstract. Large, long-lived species with slow life histories and protracted pre-breeding stages are particularly susceptible to declines and extinction, often for unknown causes. Here, we show how demographic modeling of a medium-sized raptor, the Red Kite *Milvus milvus*, can aid to refocus conservation research and attention on the most likely mechanisms driving its decline. Red Kites' survival and reproduction increased through three sequential stages for 1–2, 3–6, and 7–30 yr of age, mainly corresponding to individuals that are dispersing, attempting to gain a territory, and breeding. As typical of long-lived species, elasticities were highest for adult (≥7 yr old) survival, but this was high, with little scope for improvement. Instead, the declines were driven by an extremely low survival of pre-adults in their first years of life, which weakened the whole demographic system by nullifying the offspring contribution of adults and curtailing their replacement by recruits. For example, 27 pairs were necessary to generate a single prime age adult. Simulation of management scenarios suggested that the decline could be halted most parsimoniously by increasing pre-adult survival to the mean levels recorded for other areas, while only the synergistic, simultaneous improvement of breeding success, adult and pre-adult survival could generate a recovery. We propose three actions to attain such goals through selective supplementary feeding of both breeding and non-breeding individuals, and through mortality improvement by GPS remote-sensing devices employed as surveillance monitoring tools. Our results show how improving demographic models by using real, local vital rates rather than "best guess" vital rates can dramatically improve model realism by refocusing attention on the actual stages and mortality causes in need of manipulation, thus building precious time and resources for conservation management. These results also highlight the frequent key role of pre-adult survival for the management of long-lived species, coherent with the idea of demographic systems as integrated chains only as strong as their weakest link.

Key words: floaters; immatures; management; Milvus milvus*; non-breeders; pre-adults; pre-breeders; Red Kite; sub-adults; younger age classes.*

Introduction

Species declines and extinctions are increasing due to growing anthropogenic pressure, but understanding the causes of a decline is often challenging and fraught with uncertainty (Hoffman et al. 2010). Science should be able to guide the management response to such crisis situations in a rational, efficient, and cost-effective way (e.g., Pullin 2002, Owen-Smith 2007). To this end, demographic modeling such as population viability analysis is growingly used to forecast future extinction, identify sensitive life stages amenable to manipulation, and plan conservation action to halter or reverse a population decline (Morris and Doak 2002, Williams et al. 2002). These models are particularly useful for species with complex and protracted life cycles, such as most long-lived species, because agents of decline can potentially act and accumulate impacts over several stages of life. Some of these life stages can be difficult or impossible to observe in a direct manner (e.g., cryptic non-breeders; Rohner 1996, Jenouvrier et al. 2008), but the contribution to population change of the demographic rates attained in such phases can be often estimated (Williams et al. 2002). In turn, this allows us to (1) focus management attention on the life phases that accumulate demographic deficits driving a decline, (2) restrict the range of potential threats to more plausible ones operating during these sensitive stages, and (3) redirect conservation and research activities to diagnose the actual threats affecting key stages (Hilborn and Mangel 1997). All this may gain limited time and resources essential for management performance (e.g., Martin et al. 2012).

Long-lived raptors are ideal candidates for this approach. These species are typically large-bodied, wide-

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ranging, occur at low density, may spend eight or more years as so-called non-breeding floaters, and have low breeding rates (Newton 1979). These characteristics make them sensitive to a multitude of anthropogenic threats, susceptible to declines and extinction, and impose slow recovery rates, with population growth typically most dependent on the survival of adult life stages (Bennet and Owens 1997, Sæther and Bakke 2000, McClure et al. 2018). Birds of prey also serve important ecosystem functions and are charismatic sentinel and flagship species (Sergio et al. 2008), thus receiving marked conservation attention. Here, we use demographic modeling of a threatened raptor in an iconic national park to illustrate the usefulness of demographic models in the absence of clear knowledge of decline causes, to show the importance of detailed information on local vital rates for conservation planning, and to draw renewed attention to the strategic importance of younger age classes for the management of declining populations of long-lived species. The main objective of our analysis was to identify those sensitive life stages whose manipulation could make management most efficient.

Methods

Study species

This study focuses on the Red Kite (hereafter "Kite"), a medium-sized raptor endemic to the western Palearctic, with 95% of its world population confined to Europe (Knott et al. 2009, Aebischer and Sergio 2020). This species, typical of open landscapes with scattered woodland, has suffered dramatic declines in recent decades in all its stronghold populations of Germany, France, and Spain, which hold about 80% of the world population (Knott et al. 2009, Aebischer 2017). As a result, it is currently classed as Near Threatened in the IUCN Red List and included in Annex I of the EU Birds Directive and in Annex II of the Bern Convention on the Conservation of European Wildlife and Natural Habitats. In Spain, which holds the third largest world population, the species is classified as "at risk of extinction" since 2011 and as "at critical risk of extinction" within southern Spain (Andalucía Autonomous Region), where this study was conducted. Here, the population has declined by 95% over recent decades and is now basically restricted to Doñana National Park, whose population has also suffered a ~50% decline and is predicted by matrix demographic modeling to become virtually extinct over the next two decades (Sergio et al. 2019a). This previous demographic study was mainly based on so-called "best guess" estimates borrowed from studies on other populations. Here we update the earlier demographic model with age-dependent estimates from local individual-based data. Overall, the abovementioned widespread declines suffered by Red Kites throughout Europe are likely promoted by the sensitivity of this species to several anthropogenic risks, such as secondary poisoning, rodenticide applications, electrocution, shooting, collisions with wind turbines, and contamination (e.g., Villafuerte et al. 1998, Gómara et al. 2008, Aebischer 2009, Knott et al. 2009, Schaub 2012, Tavecchia et al. 2012, Tenan et al. 2012, Meyer et al. 2016, Blanco et al. 2017). Although these risks might be less important than in the past in some regions (Martinez-Abraín et al. 2009), the facultative scavenging habits of the Red Kite and its hunting mode of meticulously quartering the ground make this species particularly susceptible to mortality by illegal poisoned baits (Márquez et al. 2012, Tavecchia et al. 2012, Mateo-Tomás et al. 2020).

The Red Kite is a long-lived species (maximum longevity in Doñana = 30 yr), individuals are sexually mature when 1 yr old (Evans et al. 1998), but may start breeding for the first time at up to 7 yr of age (Newton et al. 1989). In these beginning pre-breeding years, young Kites often wander widely over large portions of the Iberian Peninsula, exposing themselves to the aforementioned anthropogenic threats (Tavecchia et al. 2012; authors' *unpublished radio-tracking data*).

Study area

Doñana is a 1,080-km^2 protected area located in southwestern Spain (36°56′51″ N 6°21′31″ W), it is one of the largest and most biodiversity-rich wetlands of Europe, classified as a UNESCO Biosphere Reserve, a World Heritage Site and a Ramsar wetland. It is also an iconic area for conservation action both nationally and internationally: for example, its institution sparked the creation of the World Wide Fund (García and Marín 2006). The park is world renowned for its high density of top predatory vertebrates, which represents one of the most extraordinary concentrations of Europe and contributes to the attraction of several hundred thousand visitors annually. Thus, the prospective extinction of a predatory species is considered by the park authority as a highly alarming issue in urgent need of management.

Field procedures

Red Kites were intensively surveyed since the 1970s (for details of survey and nest check procedures, see Sergio et al. [2019a]). In all analyses, we employ data collected between 1986 and 2019. Since 1986, the nestlings were marked with a white Darvic ring with a black three-character alphanumeric code, which can be read by telescope from a distance. Even though ringing activities started earlier (since 1964), it was only after 1989 that a standard protocol for systematic resighting was established. Here, we analyze 190 reencounters of 701 Kites born in Doñana Nation Park between 1986 and 2018 and marked as nestlings at the time of fledgling. As a consequence, we know the exact chronological age of all birds in the data set. Encounters included birds resighted on their territories or on carrion baits up to 2019

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included. Birds of different age or breeding status may be differentially detected by each of these methods: readings at territories mostly recorded breeders, while ring readings at carrion baits included birds of all ages and status. Because the type of resighting was recorded only since 1996, all encounter types were pooled into a single data set and analysed together. This pooling likely increased the level of recapture heterogeneity, but allowed us to include all available resightings in the analysis. For the same reason, it was not possible to separate Kite breeders from non-breeders in the beginning years of life, when both status categories can co-occur in the population for a given age. When a marked bird was detected on a territory, its nest was checked multiple times in order to ascertain the number of chicks raised to fledging age (about 55 d old; Bustamante 1993).

Modeling survival and encounter probabilities

Survival and detection probabilities, noted ϕ and p, respectively, were modeled simultaneously from observation of marked animals and estimates were derived by maximum likelihood methods using capture–mark–recapture model (Lebreton et al. 1992). We considered time (t), age (model notation follows) and their statistical interaction as predictors (factors) of survival and recapture probability. Modeling procedures should begin by assessing the fit of a general model incorporating all effects of interest and their statistical interaction (Pollock et al.1990, Choquet et la. 2009). In our case, with 34 capture–mark–recapture occasions and a relative small data set, such a model would have been over-parameterized. Therefore, we began model section from a model assuming an effect of age (11 age classes, as found for the congeneric and similar body-sized Black Kite *Milvus migrans* in Sergio et al. 2011) on survival and a full effect of time on recapture probability. This model (model 1) was considered the starting model from which we began model selection. We used the bootstrap procedure in MARK 9.0 (White and Burnham 1999) with 1,000 simulated data sets to assess the fit of this first model. More complex models, by definition, would also describe the data adequately, while the adequacy of simpler models can be assessed using model information theory (see last paragraph of this section). Model notation was complex because a given effect can be important for one age class or parameter, but not for others. We choose a model notation similar to the one used in Catchpole et al. (2000), in which each new parameter in the model was separated by a / symbol. In some models, to simplify model structure, we grouped parameters for consecutive ages into a single class. In this case, the first and the last age of the class are joined by an underscore, while a + indicates an age group that includes all ages starting from the numbered one. For example, a model assuming only two age classes for survival, one spanning the first 10 years of life and a second one from 11 onward will be denoted $\phi_{1_10}/\ \phi_{11+}$. The presence of time on a given parameter was noted in brackets. Hence, when both age classes, ϕ_{1_10} and ϕ_{11+}, respectively, were assumed to change over time the model was denoted $\phi_{1_10}(t)/\phi_{11+}(t)$. Finally, the symbol /../ will indicate that all age classes were modeled separately. Thus, our starting model, which assumed 11 age classes for survival probability and a full time effect for detection probability, was noted ϕ_1/../ ϕ_{11+}/p(t).

Because previous analyses on a sympatric population of a congeneric species with a very similar ecology and age of first breeding (the Black Kite) revealed clear age groupings of 1–2, 3–6, 7–11, and above 12 yr of age (e.g., Sergio et al. 2011, 2014), we also tested such age effects on this data set. For Black and Red Kites, these age groups correspond broadly to non-breeding individuals in their initial 2 yr of life, young adults attempting to recruit or in their first years of reproduction, 7–11 yr olds in prime age of maximum survival and breeding performance, and senescent individuals of 12 yr or older. For Red Kites, we also tested a model without the last, senescent, age class. Finally, because the differences in lifestyle during the above-described sequential stages of the life cycle could impose differences in detectability, we also tested a model with a recapture probability based on the age structure above rather than time.

Model selection was based on the Akaike's Information Criterion (AIC) corrected for small sample size (AIC_c; Burnham and Anderson 2002). The AIC is calculated as the model deviance plus twice the number of estimable parameters in the model. The AIC_c is obtained by adding a correction term calculated as $2k(k + 1)/(n - k - 1)$ where k is the number of estimable parameter in the model and n is the effective sample size. In large data sets, the AIC and the AIC_c are equivalent, but in small data sets, the correction term increases, penalizing models with a large number of parameters in relation to the sample size. In addition, we calculated the AICc weight of each model as a measure of its relative importance among those we considered (Burnham and Anderson 2002). Survival and detection probabilities, as well as their variances, were estimated by maximum likelihood procedures using program MARK 9.0 (White and Burnham 1999).

Scenarios for population managing

To explore which scenario can be more effective in managing Red Kites in Doñana National Park, we first built an age-structured demographic matrix, **M** (Caswell 2001; Appendix S1: Table S1), which included the age-dependent values of survival and fertility. The matrix **M** describes the transition of the population between year t and $t + 1$. The matrix is of dimension $n \times n$ where n corresponds to the largest number of age classes detected in the analysis of survival or fertility. The asymptotic population growth, λ, was calculated as the dominant eigenvalue of **M** (Caswell 2001). To incorporate parameter uncertainty on λ, we calculated its standard error as the

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sum of the products of each parameter sensitivity and its variance (Lande 1988). In a stable population, λ is equal to 1, while lower values indicate a declining trend. In the model we assumed no emigration of individuals born in Doñana to other populations, because none of the >700 Kites ringed in Doñana as nestlings have been resighted as breeders or recovered during the breeding period elsewhere, despite intensive searching efforts, especially in the 1990s, and much monitoring of the species throughout Spain (e.g., Viñuela et al. 1999, Molina 2015).

To investigate the potential conservation actions that could be most beneficial to halt the current decline or promote a recovery, we explored which combinations of improvements of different vital rates could generate a stable or positive population trajectory. To this end, we built 12 management scenarios (details in Table 3): each scenario was based on the improvement of one or more of the key vital rates used to build the matrix population model above, i.e., the survival and breeding performance of individuals of different ages. Improvements were planned in the following manner. (1) First, we reviewed previous studies on Red Kites that estimated the percentage decline in survival caused by anthropogenic threats (mainly poisoning and electrocution; Smart et al. 2010, Tavecchia et al. 2012, Sanz-Aguilar et al. 2015*a*). (2) We assumed that such mortality factors could be relieved through management and we simulated that the survival of each age class could be improved by the maximum percentage increase reported by such studies. If this maximum increase brought a vital rate unrealistically beyond its maximum value ever recorded in Europe (review in Sergio et al. 2019a and Table 1), we used the next, lower available estimate of percentage improvement. (3) If such improvements did not produce a positive λ, we improved the vital rate to its average European value, assuming that this could be biologically feasible by management. (4) Because no similar estimates of improvement from other studies existed for breeding performance, we improved it by multiplying it by 1.5, assuming that conservation actions such as supplementary feeding could enhance breeding performance. Larger multiplicative factors generated unrealistic estimates beyond the maximum European estimates. (5) Management scenarios were planned by improving one vital rate at a time and then by combining simultaneous improvements in multiple vital rates. (6) Finally, we also simulated the release of 5, 10, and 20 fledglings every year by increasing breeding success accordingly. In this case, we allowed breeding success to exceed the European maximum estimates. This approach assumes that the released fledglings will survive the same as wild ones, which is not always the case (e.g., Smart et al. 2010, Tavecchia et al. 2011). Because releases of 5, 10, and 15 fledglings failed to stabilize λ, we only present the scenario of 20 released fledglings. Note that this would be considered a very large release effort (Sanz-Aguilar et al. 2015*b*). Throughout, all analyses were implemented with R (R Development Core Team 2019), using the library popbio (Stubben and Miligan 2007). Values reported are means ± SE.

TABLE 1. Review of estimates of Red Kite survival by age class.

Area	Survival of age class						Reference
	1 yr old	2 yr old	≥3 yr old	1–2 yr old	3–6 yr old	7–30 yr old	
England, Midlands	0.80	0.94	0.93				Evans et al. (1999)
Scotland	0.37	0.72	0.87				Smart et al. (2010)
Wales, uplands	0.60	0.79	0.94				Newton et al. (1989)
Switzerland	0.45	0.74	0.84				Schaub (2012)
Germany	0.40†	0.74†	0.83†				Katzenberger et al. (2019)
Spain, Mallorca	0.72	0.82	0.79				Tavecchia et al. (2012)
Spain, Menorca	0.54	0.83	0.84				Sanz-Aguilar et al. (2015*a*, *b*)
Spain, Doñana	0.308	0.385	0.853	0.456	0.706	0.874	This study
Minimum European estimates‡	0.37	0.72	0.79				
Mean European estimates‡	0.55	0.80	0.86	0.676§	0.830¶	0.860¶	
Maximum European estimates‡	0.80	0.94	0.94				

Notes: European minimum, maximum, and mean rates of survival were also estimated to place into context the estimates of the Red Kite population of Doñana National Park and to explore the benefits of improving certain vital rates by management (see Methods and Table 3).

† The authors report various time-dependent estimates, the values reported here are for the 1990s, i.e., the central period of the study.
‡ Calculated without the Doñana estimates.
§ Mean of the mean European survival for 1 and 2 yr olds.
¶ Mean of the mean European survival for 2 and above 2 yr olds.
Mean European survival for individuals above 3 yr olds.

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Results

Age-dependent survival and fertility

The survival analysis was performed by considering the capture histories of 701 individuals marked as fledglings and resighted during the period 1986–2019 (effective sample size = 887). The bootstrap procedure indicated that the model $\phi_1/\phi_2/\ldots/\phi_{11}p(t)$ (model 0; Table 2) described the data adequately (deviance = 597.52; $P = 0.785$). Model selection indicated that a further regrouping of the number of age classes was possible (models 1–6; Table 2). In particular, assuming the same age structure found for a similar species, the Black Kite (see *Methods*), led to a lower AIC_c value (model 4). Similarly, changing age for time in the probability of detection further reduced the AIC_c value (model 7). The retained model assumed a structure in three age classes for both survival and detection probabilities (Table 2, model 7). According to this model, survival was low in the first two years of life, intermediate between 3–6 yr old, and levelled off after 7 yr of age (Fig. 1). Similarly, the number of young produced per territorial pair progressively increased from 0 for 1–2 yr old to 0.54 for 3–6 yr old and 0.82 for Kites above 7 yr of age (Fig. 1).

Population model and perturbation analyses

The population growth rate was estimated at $\lambda = 0.912$ (SE = 0.024; 96% CL: 0.865–0.959), a measure comparable to the mean population growth rate of 0.934 observed during the last decade. This confirms the current decline and functional extinction of the population within two decades (Sergio et al. 2019a). According to the stable age distribution, 35% of the population was composed by individuals of 1–2 yr of age, 16% by 3–6 yr old and 49% by adults, suggesting a paucity of individuals of intermediate age, i.e., of floaters available for recruitment. As expected, the elasticity of the oldest part of the life cycle (age 7+) was higher than the one for younger age classes (0.78 vs 0.22, respectively) with an even lower contribution for fertility (0.038 vs. 0.007). The sensitivity values showed an identical pattern (Appendix S1: Table S2).

In Fig. 2, we show the combination of survival and fertility values necessary to generate a stable, increasing or decreasing growth rate for each age class, while controlling for variation in vital rates of other age classes (i.e., while keeping vital rates at their mean levels for the other two age classes). At current levels of survival, there are no biologically realistic levels of fledgling production capable to produce a stable population for any of the three life stages (see details in *Discussion*).

In the management scenarios (Table 3), improving any single vital rate at a time never halted the decline. Only the combination of multiple improvements generated a positive λ. The most parsimonious action to halt the decline was the improvement of the survival of both the younger age classes to European mean values. Improvement of the survival of all age classes produced a slight 3.4% increase, but only the simultaneous improvement of all survival levels and of breeding output led to a potential recovering trend of 5–7%. The release of captive fledglings, even in large numbers (20 per year), was insufficient by itself to halt the decline.

Discussion

Red Kites in our population progressed through a sequence of life stages that were similarly outlined by survival and reproduction and were coherent with those of similar-sized, congeneric, and sympatric Black Kites (Sergio et al. 2011). These stages captured the

Table 2. Modeling age and time in survival and recapture probabilities using the encounter histories of 701 Red Kites marked as fledglings in Doñana National Park (Spain).

Model	Survival	Detection	np	Deviance	AIC_c	ΔAIC_c	AIC_c *w*
7	**ϕ1_2/ ϕ3_6/ ϕ7+**	***p*1_2/ *p*3_6/*p*7+**	**6**	**645.3369**	**1,357.9512**	**0**	**0.9999**
4	ϕ1_2/ ϕ3_6/ ϕ7+	*p*(*t*)	35	607.8219	1,381.3019	23.3507	0.0001
5	ϕ1_2/ ϕ3+	*p*(*t*)	34	611.6079	1,382.9201	24.9689	0
1	ϕ1_2/ ϕ3_6/ ϕ7_10/ ϕ11+	*p*(*t*)	36	607.5535	1,383.2064	25.2552	0
3	ϕ1/ ϕ2/ ϕ3+	*p*(*t*)	35	610.6042	1,384.0842	26.133	0
2	ϕ1/ ϕ2/ ϕ3/ ϕ4+	*p*(*t*)	36	608.8324	1,384.4852	26.534	0
6	ϕ1/ ϕ2/ ϕ3/ ϕ4+	*p*1/*p*2/*p*3/*p*4+	8	669.6571	1,386.3398	28.3886	0
0	ϕ1/. . . / ϕ11+	*p*(*t*)	43	597.5186	1,388.526	30.5748	0
8	ϕ1/ ϕ2/ ϕ3/ ϕ4+	*p*(.)	4	697.8456	1,406.4097	48.4585	0
9	ϕ(.)	*p*(.)	2	819.0062	1,523.5386	165.5874	0

Notes: The effective sample size was 887. Notation: np, number of parameters in the model; Deviance, model deviance, AIC_c, Akaike's Information Criterion corrected for sample size; ΔAIC_c, difference in AIC_c value between the target model and the model with the lowest AIC_c; AIC_c *w*, Akaike weight; ϕ, survival probability; *p*, reencounter probability. Parameter subscripts indicate Kites' age: ., constant; *t*, time; + indicates that the age group includes all ages starting from the mentioned one; _ indicates that the age group spans all the ages between the first and last one joined by the underscore. The best model (lowest AIC_c value) is highlighted in boldface type. See *Methods* for details. For conciseness of presentation, we report only the best nine models. The complete list of models is available on request from the authors.

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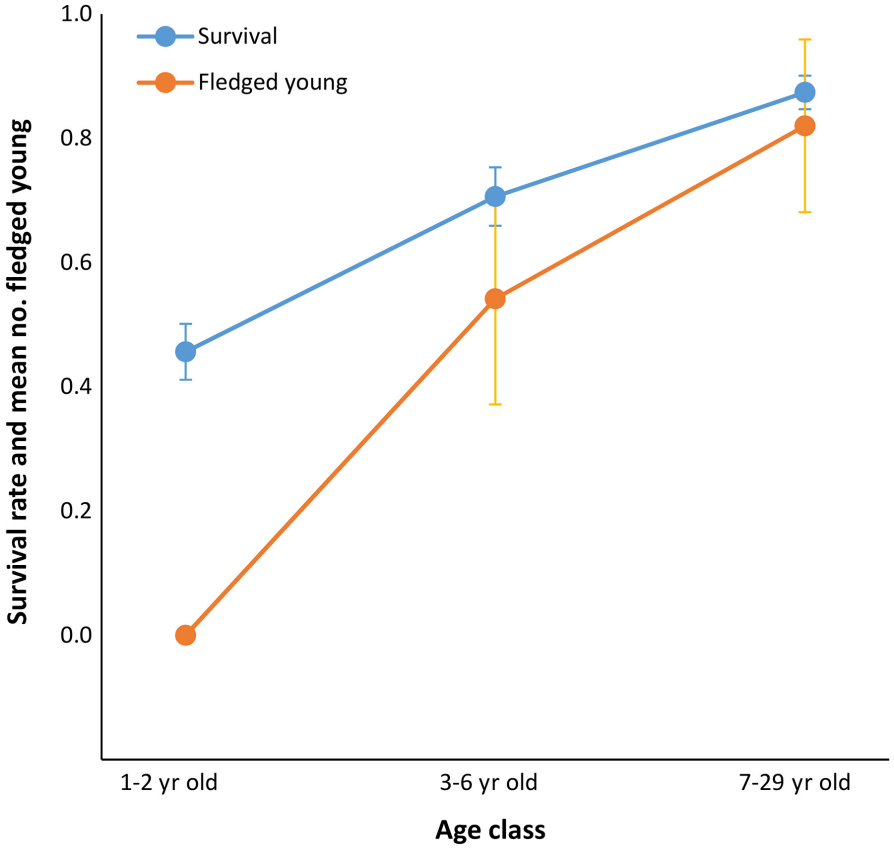


FIG. 1. Age-structured survival and breeding output by Red Kites in Doñana National Park, Spain. Bars represent SE.

differential challenges likely encountered by Kites at each step of their breeding and non-breeding careers. At 1–2 yr old, inexperienced Kites are mainly roaming over vast unfamiliar areas, exposing themselves to a large number of anthropogenic risks, such as poisoning or electrocution (Smart et al. 2010, Tavecchia et al. 2012). At 3–6 yr old, low-quality individuals have been filtered out of the population, but a portion of individuals of this age progressively starts breeding and the above anthropogenic risks for wandering floaters sum up to those of intensive fighting for access to territories and to the cost of reproduction for inexperienced breeders. Beyond 7-yr old, Kites are usually tied to a territory within the safer settings of the national park, where they attain their maximum breeding and survival rates.

Population growth reflected the contribution of the demographic processes occurring through all the above life stages. In general, analyses of population viability based on demographic parameters provide reliable information on the state of a population over the medium term (Brook et al. 2000). Also, perturbation analyses of population models, i.e., elasticity and sensitivity of demographic parameters, can help to focus conservation actions on the most sensitive parameters (Sæther and Bakke 2000; but see Mills et al. 2001 for caution on the use of elasticity). In our case, several lines of evidence suggested that the population decline was driven by problems affecting survival of the younger age classes. First, when compared to European estimates from previous studies, survival of the younger age classes (1–2 and 3–6 yr old) was 15–33% lower than the European mean and even 6–16% lower than the European minimum estimates (Table 1). If we re-estimate our survival for ages 1, 2, and above 2 yr old, to conform to the age classes of previous Red Kite studies, the survival for 1 and 2 yr olds is particularly low (44–52% and 17–47% lower than mean and minimum European values, respectively: Table 1). Second, the survival of adult age classes was above the European mean level and adequate for a raptor of this body size (Newton et al. 2016). Third, the stable age distribution suggested a paucity of young floaters available for filling territorial gaps through recruitment. For example, in an earlier study on a stable population of the sympatric Black Kite, we found a stable age distribution with similar proportions of individuals of 1–2, 3–6, and above 7 yr of age (33%, 35%, and 32% of the population, respectively). The corresponding figures were 35%, 16%, and 49% for Red Kites, suggesting population imbalance due to erosion of the floaters contingent. Fourth, in our management

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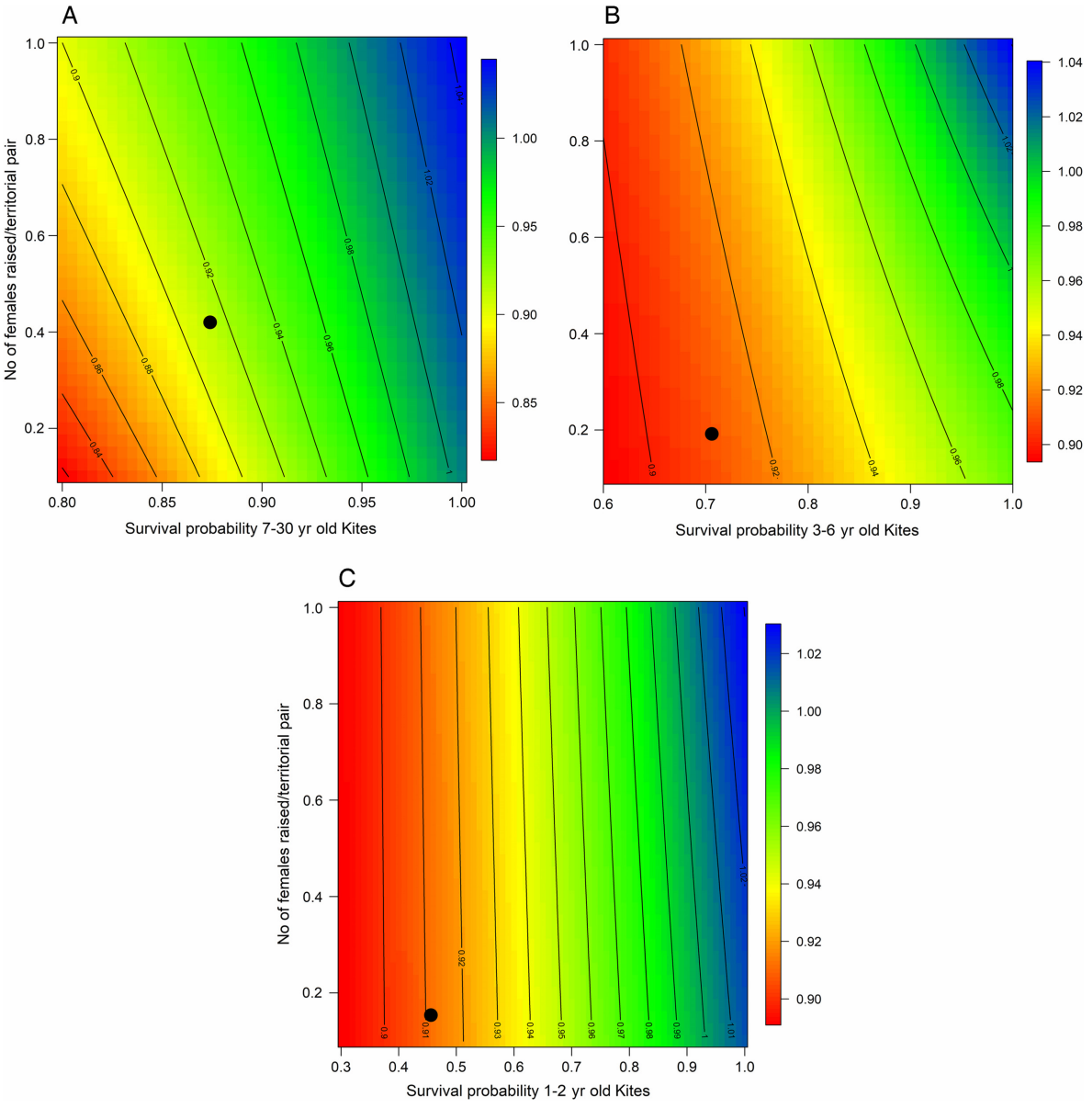


FIG. 2. Combinations of survival and fecundity values necessary to generate a stable, increasing or decreasing growth rate in Red Kites of (C) 1–2 yr of age, (B) 3–6 yr of age, and (A) 7–30 yr of age. Note that the survival–fecundity combination of each panel (i.e., for each age class) was calculated while keeping vital rates at their mean levels for the other two age classes. Thus, for example, λ-isobars for 7–30 yr old adults were calculated while keeping survival and fecundity values for ages 1–2 and 3–6 at their respective mean levels. Fecundity is the number of females raised per territorial pair. The black circles in the graph illustrate the combination of survival–fecundity estimates for the Doñana population.

scenarios, improvement of survival of the younger age classes (1–6 yr old) was sufficient to arrest the decline, which was not attained by improving adult survival alone within realistic boundaries. Finally, Fig. 2 illustrates well how the low survival of young Kites undermined the whole demographic system. Each panel of this figure shows the combination of survival and fertility that generates a certain level of λ for each age class, while keeping all the other vital rates constant at their mean values (e.g., at the locally low survival levels for young Kites). Although current levels of adult survival are adequate, panel A shows how the population would need an unrealistic breeding output >4 young/territorial pair to reach a λ of 1. This is far beyond the maximum ever recorded for the species (review in Sergio et al. 2019a). The population would need so many fledglings to attain stability because the probability that each one reaches recruitment age is so low. For example, at current levels, less than 14% of the fledglings produced would reach the mean age of recruitment of 3 yr and

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TABLE 3. Simulations of management scenarios based on theoretical improvements of the current survival and reproductive rates of the Red Kite population of Doñana National Park (see *Methods* for details).

Scenario	Rationale	Scenarios combined	Survival set to			λ	Trend (%)
			1–2 yr old	3–6 yr old	7–30 yr old		
1	increase ϕ1_2 to its European mean value†		0.676	–	–	0.951	−4.9
2	increase ϕ 3_6 to its European mean value†		–	0.830	–	0.939	−6.1
3	increase ϕ 7–30 of 5.75%‡		–	–	0.924	0.954	−4.6
4	multiply the mean number of fledged young by 1.5 for all age classes§		–	–	–	0.929	−7.1
5	release 20 fledglings every year¶		–	–	–	0.957	−4.4
6	increase ϕ 1_2 and ϕ 3_6 to European mean values	1 + 2	0.676	0.830	–	**0.996**	**−0.4**
7	increase ϕ 1_2 and ϕ 3_6 and ϕ 7–30 as above	1 + 2+ 3	0.676	0.830	0.924	**1.026**	**+2.6**
8	increase ϕ 1_2 to its European mean value and multiply FY by 1.5	1 + 4	0.676	–	–	0.983	−1.7
9	increase ϕ 3_6 to its European mean value and multiply FY by 1.5	2 + 4	–	0.830	–	0.964	−3.6
10	increase ϕ 7–30 of 5.75% and multiply FY by 1.5	3 + 4	–	–	0.924	0.968	−3.2
11	increase ϕ 1_2 and ϕ 3_6 to European mean values and multiply FY by 1.5	1 + 2 + 4	0.676	0.830	–	**1.038**	**+3.8**
12	increase survival of all age classes and multiply FY by 1.5	1 + 2+ 3 + 4	0.676	0.830	0.924	**1.063**	**+6.3**

Notes: λ, population growth rate; ϕ, survival; –, parameter of the matrix model left unchanged; FY, mean number of fledged young. Values of λ corresponding to population stability or recovery (>0.99) are highlighted in boldface type.

†Simply removing the percentage mortality caused by anthropogenic threats according to other studies produced an even lower λ (see Methods).

‡The current estimate was already close to the European population mean (see Discussion). The value of 5.75% is the survival decline imposed by poisoning in Smart et al. (2010). Survival deficits imposed by anthropogenic impacts reported in other studies (Tavecchia et al. 2012, Sanz-Aguilar et al. 2015a) led to unrealistic improvements with outcome adult survival over 1.00.

§The number of fledged young was set to 0.813 for −6-yr-old breeders and to 1.230 for breeding adults. Multiplying estimates by higher numbers (e.g., by 2) led to estimates of breeding success beyond the maximum ever reported for the species.

¶Released fledglings are added to the current average fledgling production by wild pairs. The simulation assumes that released fledglings will survive as wild ones, which is not always the case (e.g., Smart et al. 2010).

<4% would reach the prime age of maximum reproductive value of 7 yr. This implies that seven breeding attempts would be needed to generate one fledgling that reaches age three and 27 to produce a prime age adult. Furthermore, even if an individual reached such prime age, its higher breeding output would still be counterbalanced by the high mortality of its offspring. Similarly, fledgling production is currently low by European standards (Sergio et al. 2019a), but increasing it (e.g., through supplementary feeding or release of hacked young) would be worthless unless their subsequent survival is markedly increased. In this sense, the decent performance and high elasticity of adult breeders was cancelled out by the high mortality of young Kites. The offspring they produce rarely reach future generations and, when adults die, there will be progressively fewer floaters to replace them, leading to a sort of demographic breakdown and lending support to previous theoretical models (Penteriani et al. 2005, Lee et al. 2016). These results highlight the interdependence of the demographic processes that shape population change or stability (see also Mills et al. 2001), in agreement with the common notion that a chain is as strong as its weakest link.

Our results lend renewed support to the importance of pre-adult survival for population fluctuations of long-lived species (Gaillard et al. 1998, Bretagnolle et al. 2004, Ortega et al. 2009, Barraquand et al. 2013, Hernández-Matias et al. 2013, Demerdzhiev et al. 2015, Lee et al. 2016). In these species, adult survival is often high and buffered from environmental variation (Sæther and Bakke 2000). In the absence of major deficits in adult survival, pre-breeding mortality is likely to be a frequent culprit of declines, given the often wide-ranging lifestyle of younger individuals that heightens their risk exposure.

Management implications

Management recommendations based on parameter elasticities should take into account the opportunity and susceptibility to improvement of each vital rate, with recommended focus on those parameters that are farther from their maximum potential value and more realistically amenable to manipulation (Hiraldo et al. 1996, Gaillard et al. 1998, Ehrlén et al. 2001, Owen-Smith 2007). Also, considering the elasticity of a single demographic parameter might be misleading (Mills et al. 2001). In our case, adult survival had the highest elasticity, but its value was close to the value expected for this species and similar to those of stable or increasing populations. In contrast, pre-adult survival was markedly

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below its potential maximum. Breeding success was also low, but its improvement makes only sense if the weak link of pre-adult survival is simultaneously resolved (see previous section above). Thus, pre-adult survival seemed to be the prime target for management.

Previous work on other long-lived species has shown that younger individuals are often wandering over larger or different areas than adults, are subject to different sources of mortality and may respond differently to management actions. For example, adults are more affected by illegal poisoning, while pre-adults benefit more from human supplementary feeding in Bearded Vultures (Oro et al. 2008); adults are more likely to die from persecution and pre-adults by electrocution in Bonelli's Eagles (Real et al. 2001); while adult Egyptian Vultures are more susceptible to windmill collisions than pre-adults (Carrete et al. 2009). Thus, selective management of certain age classes is feasible in principle. In the specific case of Red Kites, previous studies in UK and Spain have suggested that younger individuals are more susceptible than adults to illegal poisoning and electrocution (Smart et al. 2010, Tavecchia et al. 2012, Sanz-Aguilar et al. 2015*a*). In this sense, the prime target for the Doñana population would be to identify dispersal areas and mortality agents in order to ameliorate them (e.g., through remote-tracking, retrofitting of dangerous electricity pylons in areas of high use, or legal persecution of game reserves involved in casualties caused by illegal poisoning).

Furthermore, the results of our management scenarios suggested that management actions simultaneously targeting multiple vital rates would be more useful than those directed at a single one (see similar conclusions in Sanz-Aguilar et al. 2015*b*). In particular, simultaneous improvements of breeding success and adult and pre-adult survival was the only way to reverse the decline rather than just halt it. Finally, releases of hacked young would not be desirable at the moment given the high pre-adult mortality for ethical reasons, to comply with IUCN translocation criteria, and because they would not halt the decline by themselves, even with large numbers of released Kites.

Integrating all the above, we propose a three-step management plan. (1) Implement an intensive marking program, with emphasis on young Kites, through GPS-tracking devices, used as near-real-time surveillance tools to uncover high-use dispersal areas, identify mortality risks and causes, and take action to resolve them. Remote-sensing tracking devices are growingly used for surveillance monitoring (Tavecchia et al. 2012, Whitfield and Fielding 2017, Murgatroyd et al. 2019, Sergio et al. 2019b). (2) Provide feeding stations in Doñana National Park in communal foraging areas, i.e., out of breeding territories, so as to mainly target non-breeding floaters, improve their body condition with safer food (e.g., carrion free from lead bullets) and retain them more within the national park (i.e., lower their risk exposure by increasing their permanence within the park). (3) Provide supplemental feeding at specific nests and territories in order to improve adult body condition, survival and breeding success.

Note that the benefits of actions 2 and 3 will depend on the effectiveness of action 1. Finally, supplemental feeding should be approached as a temporary, emergency tool and stopped as soon as the population attains a healthier status, in order to avoid the potentially deleterious consequences of artificial feeding (e.g., Oro et al. 2008, Robb et al. 2008).

In conclusion, note that a previous population viability model based on "best guesses" (Sergio et al. 2019a) pinpointed adult survival as the key vital rate for improvement. Here, improving such best guess model by integrating the actual demographic processes occurring in the local population allowed to refocus conservation attention from adult breeders to pre-adult kites and to risks likely occurring out of the protected area. This will likely improve the timing and cost-effectiveness of conservation action by focusing expensive tagging efforts on the young portion of the population most in need of surveillance monitoring. This highlights the importance of detailed demographic monitoring of endangered species and of locally fine-tuned models for management efficacy, rather than approximations based on borrowed estimates (see Sanz-Aguilar et al. [2015*b*] for similar results). The importance of the quality and comprehensiveness of the data used in demographic modeling of endangered species has been remarked on for over 20 yr (Beissinger and Westphal 1998), but a growing shift in conservation science from field-based studies to modeling-focused approaches (Ríos-Saldaña et al. 2018) is producing a growing share of viability analyses excessively based on best guess estimates. As we show here, this could lead to erroneous conclusions, misdirected plans, inefficient use of limited resources and a loss of management time, with potentially severe consequences in the case of populations approaching extinction (e.g., Martin et al. 2012).

Finally, our results lend renewed support to the strategic importance of the pre-adult sector of the populations of long-lived species for their management and conservation, given the frequently canalized survival of adult individuals. The wandering lifestyle conducive to risk exposure by pre-adults is likely to be common to many long-lived species, suggesting that demographic modeling exercises to fine-tune conservation, as here exhibited, may apply to a broad range of species and scenarios. These models can build essential time by restricting the range of threats to tackle, and enable a more rational start of the management process while more detailed information is collected.

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Supporting Information

Additional supporting information may be found online at: http://onlinelibrary.wiley.com/doi/10.1002/eap.2266/full

Data Availability

Data are available from DIGITAL.CSIC at http://dx.doi.org/10.20350/digitalCSIC/12614

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